

Bangladesh University of Business & Technology (BUBT)



LAB Report

Course Code : CSE 210
Course Title : Operating Systems Lab
Experiment No. : 04

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Intake: 52
Section: 01
Program: B.Sc. Engg. in CSE

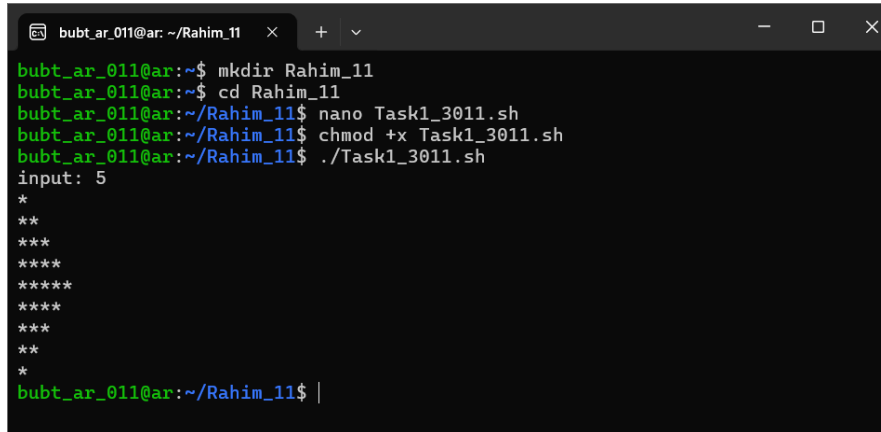
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Signature of Teacher

1. Print a bash script to generate the following pattern.

```
nano Task1_3011.sh
chmod +x Task1_3011.sh
./Task1_3011.sh
```



```
bubt_ar_011@ar: ~/Rahim_11  x  +  v
bubt_ar_011@ar:~$ mkdir Rahim_11
bubt_ar_011@ar:~$ cd Rahim_11
bubt_ar_011@ar:~/Rahim_11$ nano Task1_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task1_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task1_3011.sh
input: 5
*
**
***
****
*****
*****
****
***
**
*
bubt_ar_011@ar:~/Rahim_11$ |
```

Inside 20234103011_Info.sh:

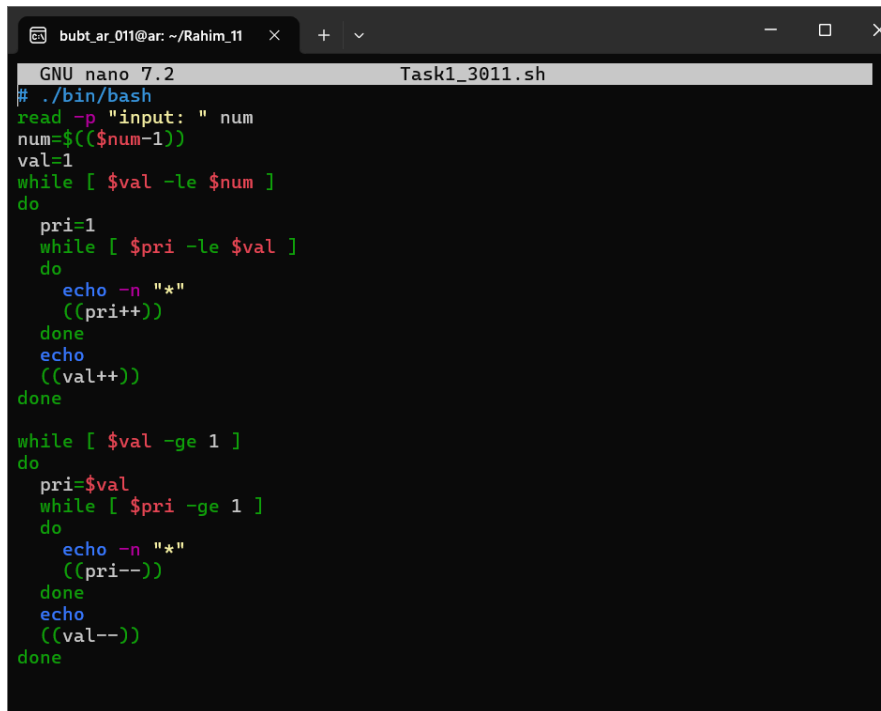
```
# ./bin/bash
read -p "input: " num
num=$((num-1))
val=1
while [ $val -le $num ]
do
    pri=1
    while [ $pri -le $val ]
    do
        echo -n "*"
        ((pri++))
    done
    echo
    ((val++))
done
```

```
while [ $val -ge 1 ]
do
    pri=$val
    while [ $pri -ge 1 ]
    do
```

```

    echo -n "*"
    ((pri--))
done
echo
((val--))
done

```



```

GNU nano 7.2 Task1_3011.sh
# ./bin/bash
read -p "input: " num
num=$((num-1))
val=1
while [ $val -le $num ]
do
    pri=1
    while [ $pri -le $val ]
    do
        echo -n "*"
        ((pri++))
    done
    echo
    ((val++))
done

while [ $val -ge 1 ]
do
    pri=$val
    while [ $pri -ge 1 ]
    do
        echo -n "*"
        ((pri--))
    done
    echo
    ((val--))
done

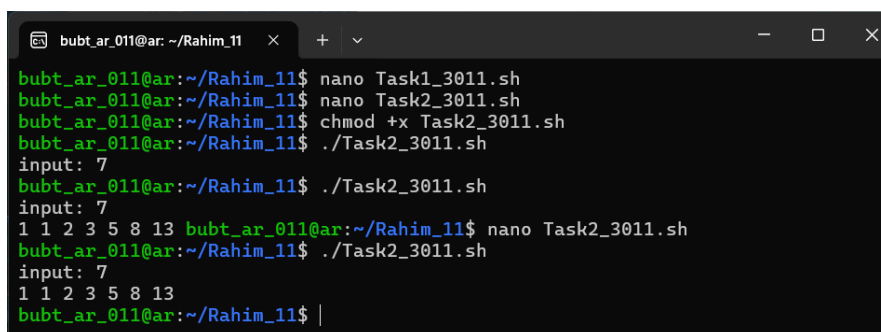
```

2. Print the Fibonacci series in terminal with a bash script.

```

nano Task2_3011.sh
chmod +x Task2_3011.sh
./Task2_3011.sh

```



```

bubt_ar_011@ar: ~/Rahim_11
bubt_ar_011@ar:~/Rahim_11$ nano Task1_3011.sh
bubt_ar_011@ar:~/Rahim_11$ nano Task2_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task2_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13 bubt_ar_011@ar:~/Rahim_11$ nano Task2_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13
bubt_ar_011@ar:~/Rahim_11$ |

```

Inside 20234103011_Operations.sh:

```

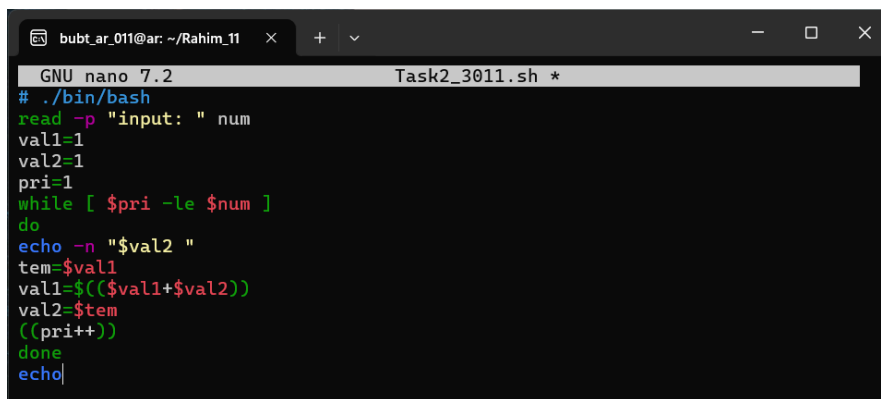
# ./bin/bash

```

```

read -p "input: " num
val1=1
val2=1
pri=1
while [ $pri -le $num ]
do
    echo -n "$val2 "
    tem=$val1
    val1=$(( $val1+$val2 ))
    val2=$tem
    ((pri++))
done
echo

```



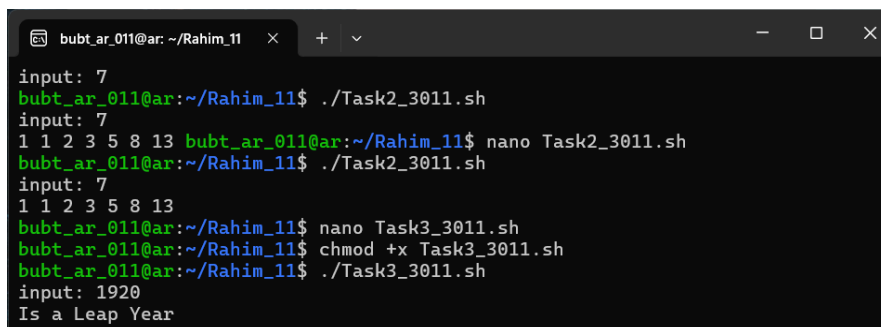
A screenshot of a terminal window with a dark background. The title bar shows 'bubt_ar_011@ar: ~/Rahim_11'. The terminal content shows the nano editor editing 'Task2_3011.sh'. The script code is visible, matching the one in the first block. The cursor is at the end of the 'echo' command.

3. Write a bash script to find if a year is a leap year or not.

```

nano Task3_3011.sh
chmod +x Task3_3011.sh
./Task3_3011.sh

```



A screenshot of a terminal window showing the execution of the script. The user enters '7' and the script outputs '1 1 2 3 5 8 13'. Then the user enters '1920' and the script outputs 'Is a Leap Year'.

```

input: 7
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13 bubt_ar_011@ar:~/Rahim_11$ nano Task2_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13
bubt_ar_011@ar:~/Rahim_11$ nano Task3_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task3_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task3_3011.sh
input: 1920
Is a Leap Year

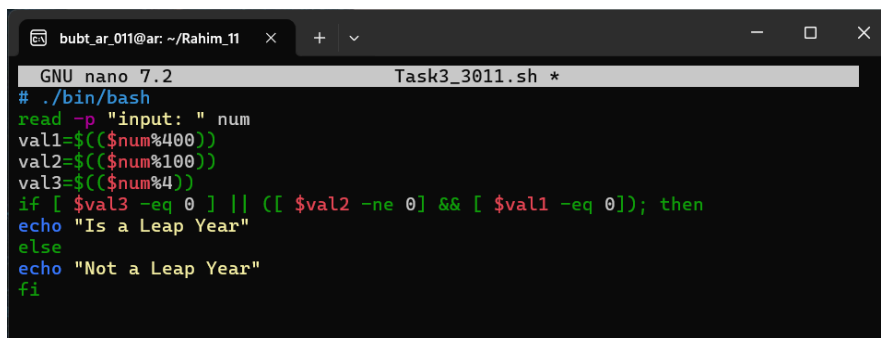
```

Inside 20234103011_AgeCheck.sh:

```

# ./bin/bash
read -p "input: " num
val1=$(( $num%400))
val2=$(( $num%100))
val3=$(( $num%4))
if [ $val3 -eq 0 ] || ([ $val2 -ne 0 ] && [ $val1 -eq 0]); then
    echo "Is a Leap Year"
else
    echo "Not a Leap Year"
fi

```



The screenshot shows a terminal window with the title 'Task3_3011.sh *'. The user is in a nano editor, creating a bash script. The script checks if a given year is a leap year using modulo operations. The script is saved and then executed, which prompts for input and outputs 'Is a Leap Year' or 'Not a Leap Year'.

```

GNU nano 7.2 Task3_3011.sh *
# ./bin/bash
read -p "input: " num
val1=$(( $num%400))
val2=$(( $num%100))
val3=$(( $num%4))
if [ $val3 -eq 0 ] || ([ $val2 -ne 0 ] && [ $val1 -eq 0]); then
echo "Is a Leap Year"
else
echo "Not a Leap Year"
fi

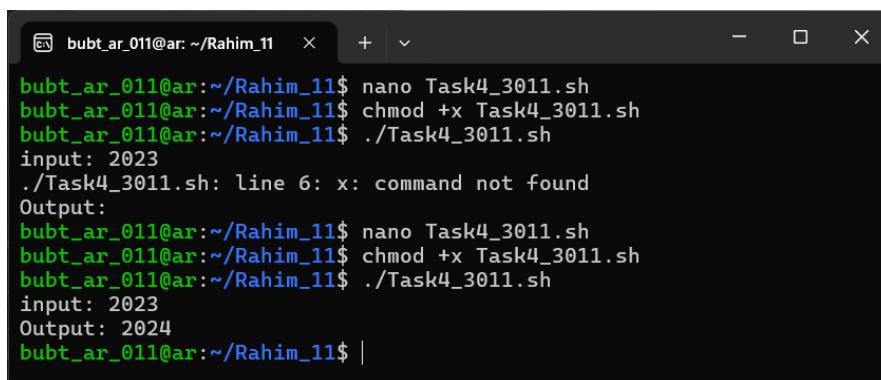
```

4. Write a bash script to find the nearest leap year from the current year.

```

nano Task4_3011.sh
chmod +x Task4_3011.sh
./Task4_3011.sh

```



The screenshot shows a terminal window where the user creates and runs Task4_3011.sh. The script prompts for input and outputs the nearest leap year. The first run with input 2023 results in an error 'command not found' on line 6, which is then corrected in the second run.

```

bubt_ar_011@ar: ~/Rahim_11$ nano Task4_3011.sh
bubt_ar_011@ar: ~/Rahim_11$ chmod +x Task4_3011.sh
bubt_ar_011@ar: ~/Rahim_11$ ./Task4_3011.sh
input: 2023
./Task4_3011.sh: line 6: x: command not found
Output:
bubt_ar_011@ar: ~/Rahim_11$ nano Task4_3011.sh
bubt_ar_011@ar: ~/Rahim_11$ chmod +x Task4_3011.sh
bubt_ar_011@ar: ~/Rahim_11$ ./Task4_3011.sh
input: 2023
Output: 2024
bubt_ar_011@ar: ~/Rahim_11$

```

Inside 20234103011_ArrayAccess.sh:

```

# ./bin/bash
read -p "input: " num

```

```

if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num %
    100)) -ne 0 ] }; then
    echo "$num is a Leap Year"
else
x   prev1=$((num - 1))
    prev2=$((num - 2))
    next1=$((num + 1))
    if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [
        $((prev1 % 100)) -ne 0 ] }; then
        echo "Output: $prev1"
    elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [
        $((prev2 % 100)) -ne 0 ] }; then
        echo "Output: $prev2"
    elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [
        $((next1 % 100)) -ne 0 ] }; then
        echo "Output: $next1"
    else
        echo "Output: $next2"
    fi
fi

```

```

GNU nano 7.2 Task4_3011.sh
# ./bin/bash
read -p "input: " num
if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num % 100)) -ne 0 ] }; then
    echo "$num is a Leap Year"
else
    prev1=$((num - 1))
    prev2=$((num - 2))
    next1=$((num + 1))
    if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [ $((prev1 % 100)) -ne 0 ] }; then
        echo "Output: $prev1"
    elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [ $((prev2 % 100)) -ne 0 ] }; then
        echo "Output: $prev2"
    elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [ $((next1 % 100)) -ne 0 ] }; then
        echo "Output: $next1"
    else
        echo "Output: $next2"
    fi
fi

```

5. Write a bash script to check whether a given number is a prime number..

```

nano Task5_3011.sh
chmod +x Task5_3011.sh
./Task5_3011.sh

```

```
bubt_ar_011@ar: ~/Rahim_11  × + ▾
bubt_ar_011@ar:~/Rahim_11$ nano Task5_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task5_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task5_3011.sh
Enter a number: 6
6 is NOT a prime number
bubt_ar_011@ar:~/Rahim_11$ ./Task5_3011.sh
Enter a number: 7
7 is a prime number
bubt_ar_011@ar:~/Rahim_11$ |
```

Inside 20234103011_ArrayAccess.sh:

```
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
    echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $((i*i)) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo "$num is a prime number"
else
    echo "$num is NOT a prime number"
fi
```

```
bubt_ar_011@ar: ~/Rahim_11  ×  +  ▾
GNU nano 7.2 Task5_3011.sh *
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
    echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $((i*i)) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo "$num is a prime number"
else
    echo "$num is NOT a prime number"
fi
```

6. Write a bash script to find and print all prime numbers between two given numbers (inclusive)..

```
nano Task6_3011.sh
chmod +x Task6_3011.sh
./Task6_3011.sh
```

```
bubt_ar_011@ar: ~/Rahim_11  ×  +  ▾
Enter a number: 7
7 is a prime number
bubt_ar_011@ar:~/Rahim_11$ nano Task6_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task6_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task6_3011.sh
Input: 11 30
11 13 17 19 23 29
bubt_ar_011@ar:~/Rahim_11$ |
```

Inside 20234103011_ArrayAccess.sh:

```
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
    if [ $num -le 1 ]; then
        ((num++))
    else
        is_prime=1
```

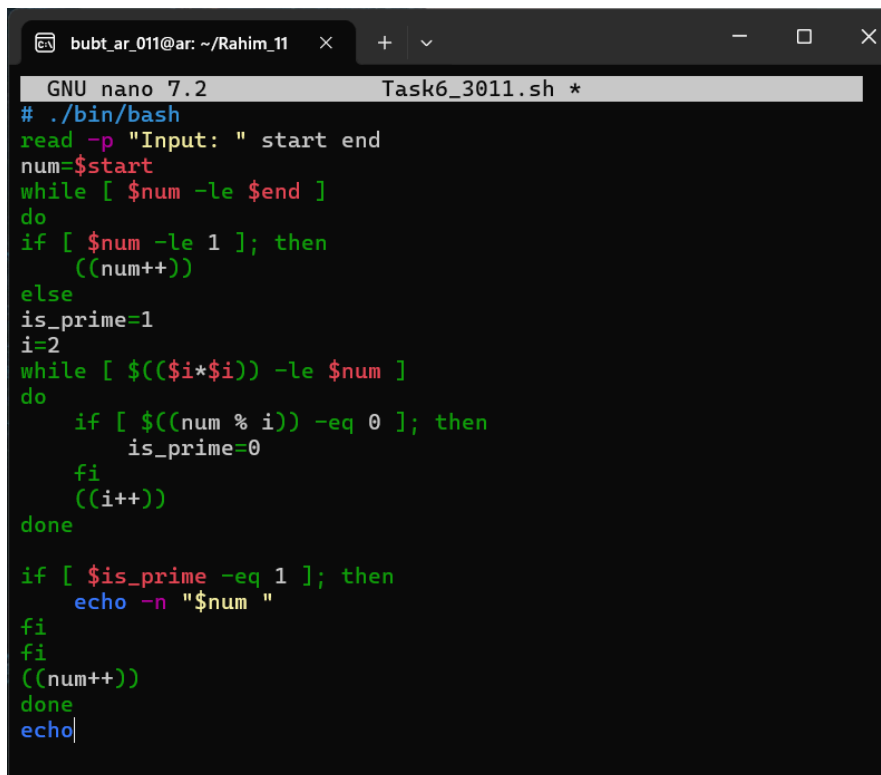


```

i=2
while [ $((i*i)) -le $num ]
do
    if [ $(num % i) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done

if [ $is_prime -eq 1 ]; then
    echo -n "$num "
fi
fi
((num++))
done
echo

```



```

GNU nano 7.2 Task6_3011.sh *
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
    if [ $num -le 1 ]; then
        ((num++))
    else
        is_prime=1
        i=2
        while [ $((i*i)) -le $num ]
        do
            if [ $(num % i) -eq 0 ]; then
                is_prime=0
            fi
            ((i++))
        done
    fi
    if [ $is_prime -eq 1 ]; then
        echo -n "$num "
    fi
    fi
    ((num++))
done
echo

```